

SKEMPI 2.0 $\Delta\Delta G$ Predictor

Antibody–Antigen Binding Affinity Change from Mutation

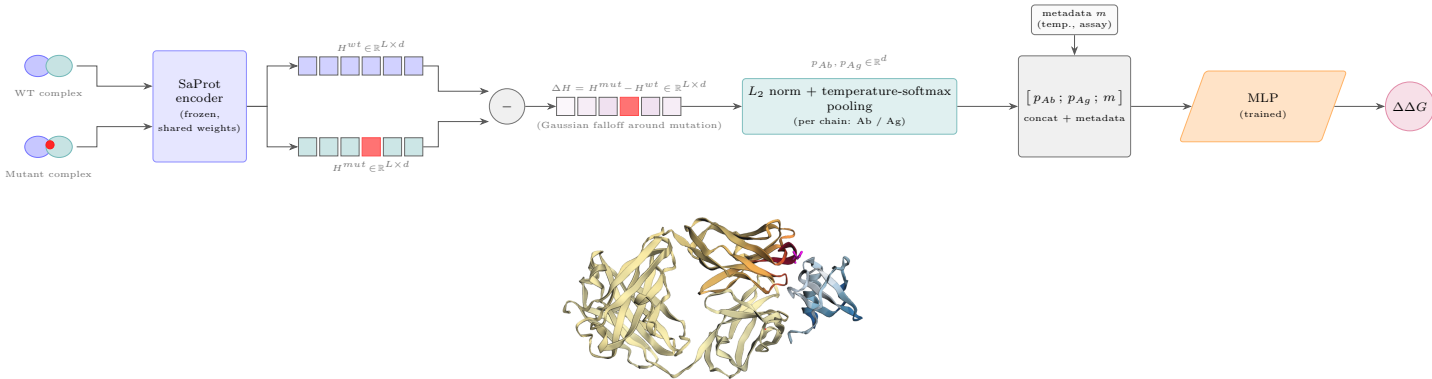
Stratification

Training and validation are split on complex+mutation identity (homology-aware, leakage-safe): the same complex with the same mutation never appears in both sets, but the same complex with a *different* mutation may — deliberately, since mutation identity (not complex identity) is the signal we want the model to learn. The validation set therefore mixes two regimes together: held-out mutations on complexes seen during training, and mutations on entirely unseen complexes. Training set: **738** bounded entries (37 complexes); validation set: **181** bounded entries across 24 complexes, of which **4** complexes are entirely absent from training and **20** also appear in training (under different mutations).

Model

Wild-type and mutant sequence/structure are passed through a single frozen pretrained structure-aware encoder (SaProt), shared between both branches, to obtain per-residue embeddings $H^{wt}, H^{mut} \in \mathbb{R}^{L \times d}$. SaProt folds sequence and structure into one amino-acid+3Di token vocabulary, so this one frozen pass already carries both modalities — no separate sequence-only encoder is used. The wild-type and mutant embeddings are subtracted position-wise, and the per-residue L_2 norm of the difference is used as an attention score in a temperature-softened softmax pool, computed *separately* over the antibody chain and the antigen chain: a plain (unsoftened) softmax was tried first and concentrated almost all weight on the mutated residue itself, starving the head of surrounding structural context, and pooling both chains together always collapsed onto whichever chain actually carries the mutation — the temperature and the per-chain split each fix one of those two failure modes. This yields two pooled “change” vectors $p_{Ab}, p_{Ag} \in \mathbb{R}^d$, concatenated with each other and with metadata features (temperature, assay type), and passed through a lightweight MLP — the only trained component in the pipeline — to predict $\Delta\Delta G$. **Loss:** plain MAE regression on the retained entries; we considered also a hinge loss for the inequality-bounded / n.b. entries, but did not use it in this model.

Rationale: we expect the residue-wise wild-type/mutant difference — computed in the local structural context around the mutation — to represent the structural change the mutation induces, and therefore to carry the information that determines what $\Delta\Delta G$ will be. This is the core assumption behind the architecture.



Example: PDB **2BDN** (human MCP-1 chemokine bound by antibody Fab 11K2), mutation **D31E** on the heavy chain. Antibody heavy chain H (orange, mutation site in red/magenta), antibody light chain L (khaki, left), antigen chain A — MCP-1 (blue, right).

Results

MAE (kcal/mol) — checkpoint `structure_saprot_only_linear_splitpool_temp5_same_pdb_allowed_20ep`, same-pdb-allowed validation split, 181 bounded rows. Baseline: per-complex train-mean $\Delta\Delta G$ (global train-mean fallback for complexes absent from training).

Breakout	MAE	Bin ($\Delta\Delta G$, kcal/mol)	n	Model	Base.
Overall (aggregate)	0.843	Large destabil. (≤ -2.0)	12	1.29	3.22
Entirely unseen complex	0.824	Mod. destabil. ($-2.0, -1.0]$	16	0.56	2.52
Same complex, new mutation	0.871	Near zero ($-1.0, 1.0$)	103	0.75	0.98
Mutation on antibody side	0.774	Mod. stabil. ($1.0, 2.0$)	27	0.74	0.50
Mutation on antigen side	0.922	Large stabil. (≥ 2.0)	23	1.34	1.68
		Overall	181	0.84	1.28